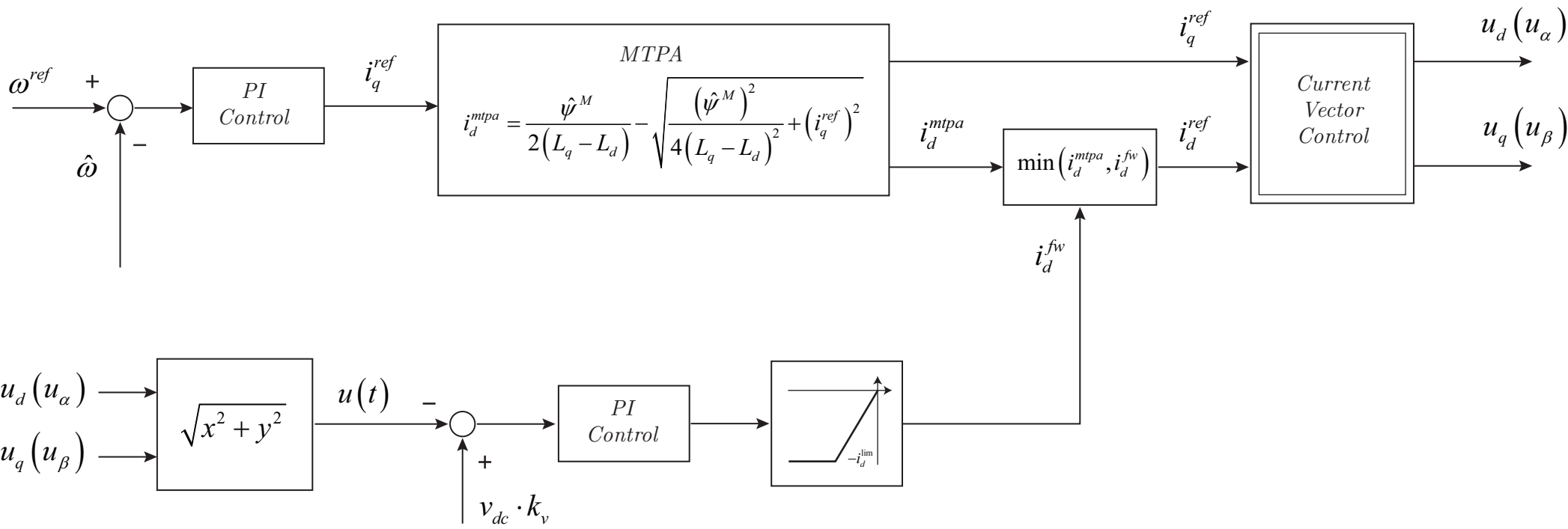
ω^{ref}
+
−
$\hat{\omega}$
PI Control
i_q^{ref}
MTPA
$i_d^{mtpa} = \frac{\hat{\psi}^M}{2(L_q - L_d)} - \sqrt{\frac{(\hat{\psi}^M)^2}{4(L_q - L_d)^2} + (i_q^{ref})^2}$
i_q^{ref}
i_d^{mtpa}
$\min(i_d^{mtpa}, i_d^{fw})$
i_d^{ref}
Current Vector Control
$u_d(u_\alpha)$
$u_q(u_\beta)$
i_d^{fw}
$u_d(u_\alpha)$
$u_q(u_\beta)$
$\sqrt{x^2 + y^2}$
$u(t)$
−
+
$v_{dc} \cdot k_v$
PI Control
$-i_d^{\lim}$